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Introduction

The Eddystone 1002 is listed as a professional grade broadcast receiver that was used as a 'comfort' set by the military during the 1970's. It was in production from 1972 to 1977. It covers 150 kHz to 30 MHz in six bands plus 88.0MHz to 108.0MHz FM.



It is an AC/DC set that sports an internal NiCad battery pack, a telescopic whip, stereo FM reception and includes binding posts for various LW/MW/SW aerial configurations and a Hi-Z recorder output. This 1002/2 seems to be an economy model and does not have the internal battery pack, telescopic whip, only receives mono FM, and is missing the aerial binding posts. In their place there is an N connector for the six AM bands. Also included is a 600 ohm line out circuit.

This particular set was a bit of a mess. The case was chipped, scratched and spattered with paint; the grey enamel on the front panel had been chipped and worn away in places. The S meter was missing the front cover, the speaker grille was missing, the tuning vernier was scraping against the dial glass and was scratched for almost 360 degrees probably as a result of the damaged speaker grille. Also missing was the DC connector and some knurled nuts from the binding posts on the back panel.

As soon as I brought the set home I took a better look at it. The set has a mains connector that is foreign to me so I found the internal 12V connections and gave the set a brief test. A slight hiss, two good dial lamps and the battery test function for the meter worked. Other than that it seemed the radio was dead as a dodo.

I put it aside and made enquiries regarding the missing components. Ian Nutt provided a NOS expanded mesh speaker grille and knurled binding post nuts

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and Chris Pettitt provided a replacement meter. When the parts arrived I stored them inside the radio until I had time to work on it.

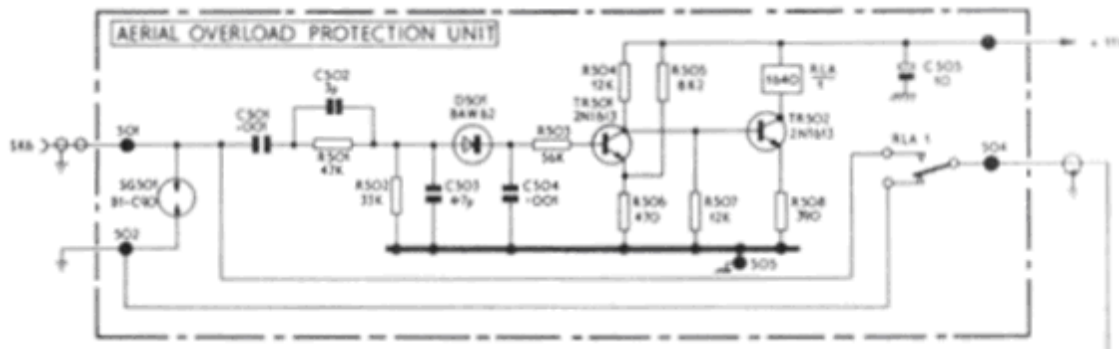
Repairs.

The first thing I found was a cable hot-wired across the mains socket then cut off and shoved back into the innards of the supply. Worst of all the cut ends were not insulated in any way! I was lucky that during my initial testing I didn't have the right mains connector or things could have turned out quite differently. It goes to show how important it is to very carefully check AC/DC radios BEFORE you start working on them or connect them to a power source.

The cable was removed and the mains supply returned to original condition. Due to the unresolved issue of the strange AC power connector the rest of the testing was carried out using a 12V DC supply. The current drain is quite modest at around 70mA during normal listening rising to around 200mA when the dial lights are switched on.

The front panel was removed and the new speaker grille and meter installed. I also reset the mechanical alignment of the dial which included moving the vernier dial back 1mm so it didn't scrape against the dial glass.

The 1002/2 incorporates a protection unit for the HF receiver. If a large transient or signal is sensed by the circuit RL1 will change over disconnecting the aerial and shorting the front end of the receiver to ground.



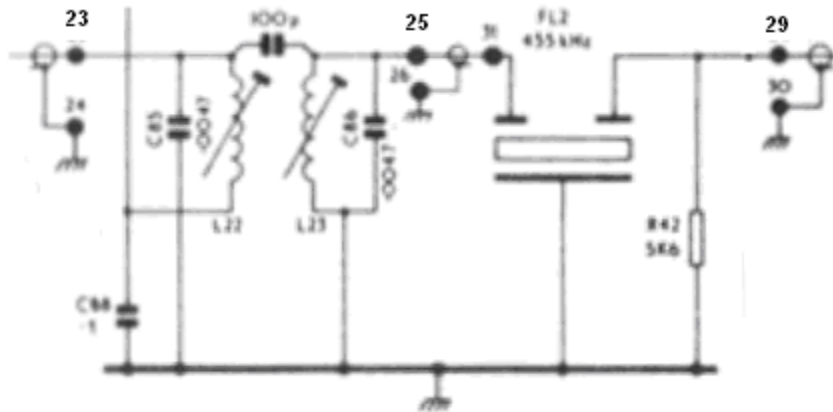
By today's standards the relay is an unusual design and replacements are not easily obtainable from hobbyist stores. In my set the circuit had been bypassed and further examination showed that the relay contacts were unreliable even after cleaning. No wonder it was disconnected. I didn't bother repairing it.

After my initial testing I thought the radio was dead but in fact it was just deaf. I found it could hear a -10dBm signal fed into the aerial socket but nothing could be heard below -50dBm. The frequency calibration was good on most bands so it

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seemed that the loss in sensitivity had a common cause and was most likely somewhere in the IF section.

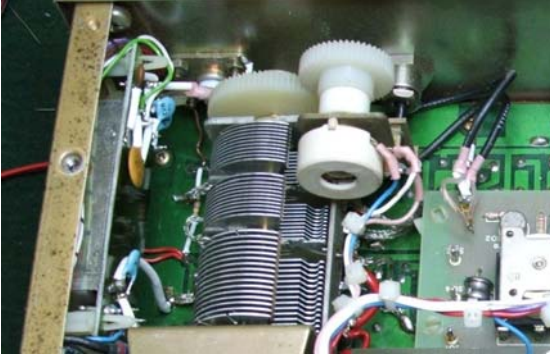
I set my signal generator to 455 kHz and connected a 'scope probe to the output. This made it easy to inject a signal into various places in the receiver and quickly zone in on the faulty area. With reference to the circuit, the radio made a big healthy noise when the signal was injected at point 29 yet was very quiet when the same signal was injected at point 23 and didn't get any better when I moved to point 25. A gentle push on the top of the ceramic filter FL2 and the radio got much louder. The reason for this change was a dodgy solder joint on the filter and once re-soldered the set then came back to life and behaved as expected.



I connected an aerial and was greeted with many MW stations of good strength. Now that I could hear signals I ran through the receiver alignment and discovered more problems. The core in the OSC coil L17 for band 3 (3.6 to 8.5MHz) was cracked and try as I might I could not move it. As a result the dial calibration at the low end of the band was out by 200 kHz.

I also found that band 4 (1.5 to 3.8MHz) OSC could not be aligned correctly and it was only by mistake that I adjusted the wrong core and found that the OSC coils for band 4 and band 5 (MW) were physically transposed compared to the diagram of the coil box in the service manual. The manual makes no mention of this and it took me a while to figure out what was going on.

Regardless of the coil transposition I still couldn't get band 4 to work correctly. The tuning range was restricted and erratic and just touching the coil would cause the generator signal to disappear. A check of the PCB foil side showed that someone had been there before and although they'd tried to make good the solder joints were pretty poor. Once I re-soldered the joints I found that band 4 was now completely dead! Once again I returned to the PCB looking for bad joints but finding none I decided to pull the OSC coil out for a closer inspection. It was here that I hit the jackpot. When the coil was removed I found one of its pins folded underneath itself - the pin had not gone through the board! I expect that this was caused by the well meaning previous owner and once the pin was straightened and the coil installed band 4 tuned up as expected.



I then turned my attention to the FM section and found the dial calibration was terrible. The FM tuner in the 1002 is tuned by a pot that is fitted with a geared wheel which runs against a larger gear attached to the AM tuning gang. The FM alignment instructions tell you to check that the pot is at its end stop when the main dial is at its limit of travel. This was not the

case with my set and I felt this was the reason for the poor calibration. Alas, once I adjusted the pot and fired the set up I found the FM dial calibration was still seriously in error. The next step was to follow through with the rest of the FM alignment. Sadly the OSC and AE cores in the FM tuner section were broken. Here then was the reason why the calibration was out and probably why the tuning pot was mis-aligned. The cores have a tongue rather than a slot in the top and require a special alignment tool to adjust them. The tops of the cores were now flat and there was no way to remove what was left of them. The end result is that the upper 10MHz of the FM band was not tunable.

Another email to Ian Nutt brought success and replacements for the dial, FM tuner (8724P) and Band 3 oscillator coil (L17) were soon on their way.

The band 4 OSC coil went in without a hitch. As it was for the other bands, the dial calibration was excellent and now the HF part of the radio was working as it should.

It wasn't so easy with the FM tuner. Ian provided me with a tuner that he thought may have been swapped out or re-furbished. He did warn me that he couldn't guarantee it was functional. Regardless, it was that or no FM at all.

Sure enough, the slug for the OSC coil in the replacement unit was broken. Additionally, the tuner in my radio was a later revision and had several extra components added to the carrier board.

I noticed that the MIX and RF coils were the same physical construction, same number of turn, and simply had an extra winding. I figured it was unlikely that Eddystone would have used a different ferrite mix for the slugs. Since I was unsure of the state of the replacement unit I thought it best to replace the coils in my unit with those from the replacement unit.

A delicate task as the PCB quality was not that good and the tracks could not handle much heat. I removed the coils and pulled the top winding off one of the coil to use for the OSC. The other coil went in without modification.



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With care and just enough heat to make a good solder joint I was able to rebuild the tuner. Thankfully my hunch about the ferrite cores was correct and the tuner performed as expected. Calibration was straight forward and even before I finished the job I knew I was on a winner. There was enough leakage to hear FM stations as I tuned back and forth along the dial.

Mains power

The remaining technical task is to see if I can find a connector to suit the mains input. I think this might be a Belling and Lee type connector but my searching has turned up nothing so far. I thought about installing a standard I.E.C. socket but there is not much room inside to accommodate something that size.



Should someone know what this connector is and where to find one I'd be pleased to hear from you.

Summary

The 1002 is general coverage receiver with solid performance that is suitable for everyday use. Whilst the lack of a BFO reduces it's versatility it is certainly a worthy addition to any collection. My 1002/2 now takes pride of place in my study and is used every day.

References

All information for this set was found on the Eddystone User Group website.

<http://www.eddystoneusergroup.org.uk/>

Some editions of the EUG Newsletter make reference to this receiver.

43-p24	FM drift found to be caused by poor solder joints and dirty trimpot in DC converter for FM tuning.
46-p13	Request for info on TAA-300 audio IC.
49-p9, 27	Advertising.
61-p11	Noisy reception caused by back to back diodes in aerial circuits.
63-p13	Editorial comments.
70-p18, 37	Editorial and description by Ted Moore.
77-p14	Editorial and troubleshooting description by Ted Moore.
86-p10, 17	Editorial and description by Ted Moore.
94-p44	No AVC - replaced IC with BC182
95-p28, 29	Ebay sales

Spare parts.

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